

Experiment	Insitu Density tests
Objective	To determine the insitu or field density of a compacted soil using; 1. Core cutter method 2. Sand cone method
Apparatus	<u>Core cutter test</u> Core cutter with detachable collar Rammer, Straight edge, small pick, spoon <u>Sand cone test</u> Sand cone apparatus Dry, clean and non-cemented sand Balance readable to 1g, drying oven

Theory

Soil compaction is used to densify a soil by reducing the void space, or the amount of air, between soil particles. In other words, soil compaction occurs when soil particles are pressed together to reduce the space between them.

The dry density of the compacted soil or pavement material is a common measure of the amount of compaction achieved during construction. It can be calculated from the field density and field moisture content data;

$$\gamma_{dry} = \frac{\gamma_{bulk}}{(1 + \omega)}$$

γ_{dry} = dry density of the compacted soil
 γ_{bulk} = field density of the compacted soil
 ω = field moisture content

- In-situ density is widely used to control the field compaction of earthworks and pavement layers.
- In-situ density of the soil enables the estimation of the soil-bearing capacity, evaluation of the pressure on underlying strata, and computation of the settlement and stability of a natural slope.

Procedure

Core Cutter Method

- Weigh the cleaned core cutter.
- Prepare the soil surface to be tested.
- Place the core cutter with collar on top of the prepared surface and drive in to the soil with the help of rammer.
- Dig the surrounding soil and take out the core cutter with soil sample.
- Level both ends and weigh the core cutter with sample.
- Remove the soil sample from core cutter, mix thoroughly and take a representative sample and determine its moisture content.

Sand Cone Method

Determination of bulk density of sand

- Weigh the empty apparatus and place it on a firm level surface and close the valve.
- Fill the cone with sand.
- Open the valve while keeping the cone at least half full of sand.
- Fill the apparatus with sand and close the valve.
- Remove excess sand from the cone and weigh the filled apparatus.

Determination of sand weight required to fill the cone

- Invert the apparatus on a clean, level surface with the valve closed.
- Open the valve and keep until sand stops running.
- Close valve and weigh the apparatus with remaining sand.
- Replace the sand removed and close the valve.

Determination of the bulk density of soil

- Prepare the surface of the location to be tested so that it is a level plane.
- Seat the inverted apparatus on the prepared plane surface over the special plate provided and mark the outline of the hole.
- Dig the test hole inside the mark being very careful to avoid disturbing the soil that will bound the hole. Soils that are essentially granular require extreme care. Place all loosened soil in a container being careful to avoid losing any material.
- Seat the apparatus in the previously marked position, open the valve and after the sand has stopped flowing, close the valve.
- Weight the apparatus with remaining sand and determine the weight of sand used in the test.
- Weight the material that was removed from the test hole.
- Mix the material thoroughly and take a representative sample for moisture determination.

Calculations - Calculate bulk density, moisture content and dry density of soil.

Results - Dry Density of the soil

Conclusion - Discuss how the calculated dry densities are used to check the specified compaction.

OBSERVATIONS

In-situ Density Test

Name

Reg Number

Course

Date

Core cutter Method

Determination of bulk density of soil

Volume of core cutter (cm ³)	
Mass of core cutter (kg)	
Mass of soil + core cutter (kg)	
Mass of soil (kg)	

Determination of the moisture content

Can number		
Mass of wet soil + can (g)		
Mass of dry soil + can (g)		
Mass of can (g)		

OBSERVATIONS

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Sand Cone method

Determination of bulk density of sand

Volume of the bottle (cm^3)	
Mass of the bottle (kg)	
Mass of the bottle filled with sand (kg)	
Mass of bottle and sand after inverting on a flat surface (kg)	
Mass of bottle and sand after filling the hole (kg)	

Determination of bulk density of the soil

Mass of Pan (kg)	
Mass of soil and pan (kg)	
Mass of soil (kg)	

Determination of the moisture content

Can number		
Mass of wet soil + can (g)		
Mass of dry soil + can (g)		
Mass of can (g)		